
Contextual information extraction of customer data from sales audio calls using a LoRA-fine-tuned language model

Thomas Hoang¹

¹Student ID: A1984077

Abstract

Sales agents on our CRM platform (uCall) currently fill in customer fields by hand after every call. This project presents a three-stage pipeline to automate that process: (1) Silero VAD removes silence and noise from the raw audio, (2) a wav2vec 2.0 model transcribes the remaining speech in Vietnamese, and (3) a LoRA-fine-tuned LLM extracts structured CRM fields from the transcript. The full staging export comprises 13,232 manual sales calls; all reported VAD, ASR, fine-tuning, and extraction results use a subset of 600 calls, with an 80/20 train-test split for the extraction stage. Evaluation uses Word Error Rate for ASR and field-level metrics on the held-out test set (JSON validity and agreement with teacher labels).

Keywords keyword1, keyword2, keyword3, keyword4

Abbreviations abbreviation1, abbreviation2, abbreviation3, abbreviation4

Significance statement

Research reports require a significance statement of between 50 and 120 words. Abbreviations are permitted, but citations cannot be included. If required, un-comment this element in the template to include. The heading is included automatically.

Scope and assumptions agreement (draft placeholder—remove later)

This section is for supervisor/student alignment only. It is not part of the final paper. Delete everything from here through the horizontal rule below once scope, data, methods, and evaluation are agreed.

Purpose

Use this page to confirm what the manuscript will claim before drafting each section. Tick or edit each item; unresolved items stay as TBD.

Problem and system context

- ☐ **Application:** uCall CRM / telesales outbound calls; manual post-call field entry is the pain point.
- ☐ **Goal:** audio → structured CRM JSON (sector, needs, interest, busy, callback time, etc.).
- ☐ **Language / domain:** Vietnamese telesales audio; stereo agent (L) / customer (R).

Data

- ☐ **Full corpus:** 13,232 manual calls (duration > 5 s) from internal staging export.
- ☐ **Experiments in this paper:** 600 calls (MAX_CALLS); rationale for subset documented.
- ☐ **Train/test:** 80/20 split after Gemini labeling (seed=42); train ≈480 / test ≈120 calls.
- ☐ **TBD:** any public release of audio or transcripts? (yes / no / anonymised subset only)

Pipeline (three stages)

- ☐ **Stage 1:** Silero VAD, 16 kHz, both channels, checkpointed segments.
- ☐ **Stage 2:** deployed Vietnamese ASR (wav2vec 2.0-based); merge to Agent: / Customer: dialogue.
- ☐ **Stage 3:** Gemini teacher labels → LoRA fine-tune (Qwen3.5 family); inference on held-out test only.
- ☐ **TBD:** report WER on how many calls / with what reference transcripts?

Models and training

- ☐ **Teacher:** Gemini 3.1 Flash Lite for response labels.

- ☐ **Student(s):** Qwen3.5-2B and/or 0.8B LoRA (list final models in paper).
- ☐ **TBD:** compare base vs LoRA? compare Gemma or other baselines?
- ☐ **Training data file:** finetuning_dataset_train.jsonl only (no test leakage).

Evaluation metrics (report on test split)

- ☐ **ASR:** Word Error Rate (WER) — if reported, define reference source.
- ☐ **Extraction:** Valid JSON %; customer_busy accuracy; sector match; schedule match; MAE for interest/rating.
- ☐ **TBD:** Precision / Recall / F1 still required, or field-level metrics above are enough?
- ☐ **Deployment:** mention FastAPI /extract demo? (in paper / appendix / omit)

Manuscript outline (agree before writing)

- ☐ Introduction — problem, pipeline summary, contributions (to write after agreement).
- ☐ Data & methodology — dataset SQL filter, VAD/ASR/LLM details, figures from notebooks.
- ☐ Results — tables from evaluation_results.csv; honest limits (subset size, teacher bias).
- ☐ Conclusion — what worked, what to scale to full 13k corpus.

Open questions for supervisor

1. Confirm 600-call subset is acceptable for the assignment vs scaling claims.
2. Confirm Gemini-as-ground-truth is acceptable or human audit on a sample is needed.
3. Confirm final page limit, citation style, and whether significance statement is required.
4. **TBD:** _____
5. **TBD:** _____

Agreed by: Student _____ Supervisor _____
 _____ Date _____

End of placeholder section — remove everything above this line before submission.

1. Introduction

1.1. Problem Statement

1.2. Motivation

1.3. Contributions

2. Related Work

2.1. Novelty and Significance

3. Methodology

4. Experimental Setup

4.1. Data

4.2. Hardware and Software

4.3. Hyperparameters

5. Results

6. Discussion

6.1. Obstacles and Strategies

6.2. What We Learned

6.3. Future Directions

7. Conclusion